

Post-quantum secure messaging layer for defense tactical communications.

Defense × Tactical and mission-critical communications × Post-quantum cryptography · OS145 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
81 is the world moving	42 can we play, can we win	MEDIUM 1 named in the evidence	€95.9m observed evidence · per year	NEXT regulation adopted or propose...	L1 13 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 0.0 published reference(s) in Defense, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

This opportunity is a post-quantum secure messaging layer for defense tactical communications, addressing the threat of harvest-now, decrypt-later attacks. Defense organizations need to protect sensitive communications now, before quantum computers can break current encryption. Orange Business can deliver this by combining its cybersecurity and defense expertise with its Quantum Defender offer, leveraging its sovereign cloud and certifications.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Observed: tendered contract value, annualised	€95.9m	€95.9m	€1.2m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

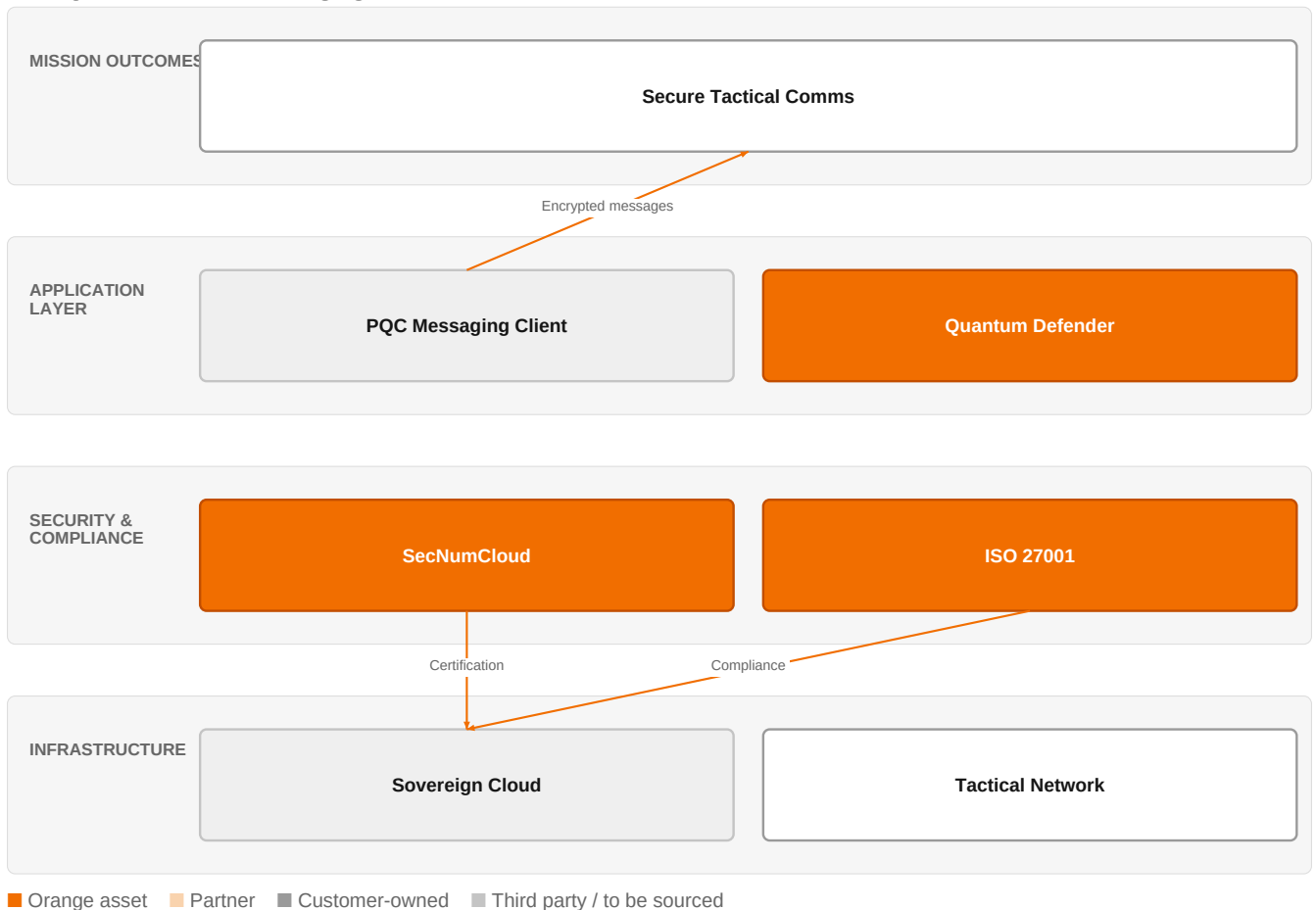
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€95.9m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-21..2026-08-05
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 5 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

Post-Quantum Secure Messaging for Defense



This diagram shows how Orange's Quantum Defender and sovereign cloud infrastructure deliver a post-quantum secure messaging layer for defense tactical communications.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

The market is shifting due to the imminent threat of cryptographically relevant quantum computers (CRQCs) that can break RSA and elliptic-curve cryptography, as highlighted in multiple studies. Adversaries can already record encrypted traffic today to decrypt later, a temporal risk that makes current communications vulnerable. Regulatory and strategic moves, such as the Department of War's enterprise post-quantum cryptography strategy to insulate defense networks by 2031, are pushing defense organizations to act. However, a critical gap remains in operational models to quantify and prioritize exposure, especially for embedded systems in tactical communications.

Sources: SIG-40B64E273856 International Journal 2026-07-30; SIG-13AAE2CD702E MDPI AG 2026-01-19; SIG-5C7568EBD975 Elsevier BV 2026-01-01; SIG-CA65D28B8AC6 openalex.org 2025-12-18; SIG-FB099EF70AC5 Quantum Computing Repo 2026-07-01; SIG-D45704585F54 Elsevier BV 2026-01-01

Who buys, and why now

The buyer is typically a defense procurement officer or CISO within a defense ministry or prime contractor. The pain is felt by tactical communications officers who rely on secure messaging for mission-critical operations, fearing that their encrypted communications could be decrypted in the future. The trigger is often a new mandate or strategy, like the Department of War's 2031 deadline, or a security audit that reveals the vulnerability of current cryptographic assets.

Why it is hot now — every claim bound to a dated source

- The existence of a post-quantum E2EE layer for messaging indicates the feasibility of deploying quantum-resistant encryption in communication tools. [SIG-D352790E1A52]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution using its Cybersecurity experts and Defense and security experts to conduct a cryptographic asset discovery and inventory, identifying all communication endpoints. Then, leveraging Orange Group researchers, it would design a post-quantum secure messaging layer, integrated with its Quantum Defender / quantum-safe offers. The solution would be deployed on sovereign infrastructure, certified with SecNumCloud and ISO 27001, ensuring compliance and trust.

Why Orange can win

Orange can win because it combines deep defense expertise with cybersecurity capabilities and a sovereign, certified cloud offering. The SecNumCloud certification is a differentiator for defense clients requiring sovereignty. However, the assets are thin: there is no specific post-quantum messaging product, so Orange would need to build on its Quantum Defender offer and research capabilities, which may be less mature than competitors' dedicated PQC solutions.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Quantum Defender / quantum-safe offers	offer	L1	offer provides the technology in a matching domain	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
SecNumCloud	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
Cybersecurity experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 6.3; 1 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitors include Thales, Airbus Protect, and Atos/Eviden, all active in defense and selling post-quantum cryptography. Microsoft, though a partner, is also a competitor with strong enterprise reach. IBM and Entrust offer PQC solutions, while ID Quantique specializes in quantum-safe security. Dassault Systemes/3DS Outscale provides sovereign cloud, which could be an alternative for hosting. Customers will compare Orange's sovereign cloud and defense expertise against these specialists.

Sources: SIG-5C7568EBD975 Elsevier BV 2026-01-01

Competitor	Category	Position here	Basis	Evidence
Microsoft	Hyperscaler	named in this topic's evidence — also an Orange partner	evidenced	SIG-5C7568EBD975
Atos / Eviden	Global systems integrator	sells Post-quantum cryptography; active in Defense; competes in Cybersecurity	structural	—
IBM	Global systems integrator	sells Post-quantum cryptography; competes in Cybersecurity	structural	—
Airbus Protect	Security specialist	sells Post-quantum cryptography; active in Defense; competes in Cybersecurity	structural	—
Entrust	Security specialist	sells Post-quantum cryptography; competes in Cybersecurity	structural	—
ID Quantique	Security specialist	sells Post-quantum cryptography; competes in Cybersecurity	structural	—
Thales	Security specialist	sells Post-quantum cryptography; active in Defense; competes in Cybersecurity	structural	—
Dassault Systemes / 3DS Outscale	Sovereign cloud provider	active in Defense	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 What is your current cryptographic asset inventory for tactical communications, and have you identified which systems are vulnerable to harvest-now, decrypt-later attacks?
- 2 Do you have a mandated deadline for post-quantum migration, such as a national strategy or defense directive?
- 3 What are your sovereignty requirements for hosting and key management in tactical environments?
- 4 How do you currently manage secure messaging across different echelons, and what are the interoperability constraints?
- 5 Have you already evaluated any post-quantum solutions, and what were the gaps you identified?

Objections you will meet

They will say	You can answer
We already have a PQC strategy in place with another vendor.	That's good to hear. However, our approach focuses on the specific tactical messaging layer and leverages our sovereign cloud and defense expertise, which may complement your existing strategy. We can help you assess gaps in your current plan.
Post-quantum cryptography is not urgent; quantum computers are years away.	While large-scale quantum computers may be years away, the harvest-now, decrypt-later threat means your encrypted communications today could be decrypted in the future. Defense communications have long confidentiality requirements, making this urgent now.
We are concerned about performance and interoperability in tactical environments.	That's a valid concern. Our solution is designed with defense experts to minimize latency and ensure interoperability with existing systems. We can pilot it in a representative environment to demonstrate performance.

Risks and unknowns

The main risk is that the technology is still evolving; standards for PQC are not fully finalized, and the operational impact on tactical communications is unknown. There is also a risk that defense clients may prefer specialized PQC vendors with proven products. Unknowns include the timeline for quantum computers and the specific regulatory requirements that will apply.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a defense tactical communications tender, differentiating our approach by highlighting our ISO 27001 and SecNumCloud certifications and the potential of our Quantum Defender offers, while outlining how our cybersecurity and defense experts can support implementation.
Sales	In a conversation with a defense ministry's head of communications, say: 'We are exploring post-quantum secure messaging for tactical communications, and we'd like to understand your requirements for quantum-resistant encryption in mission-critical scenarios.'
Strategist / Innovator	Commission a deep-dive brief on the feasibility of a post-quantum E2EE messaging layer for tactical defense communications, leveraging Orange Group researchers and cybersecurity experts, and assess the regulatory timeline for adoption.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-40B64E273856	2026-07-30	International Journal of ...	1	Post-Quantum Cryptography Migration for Enterprise Security
SIG-B8561992126B	2026-07-30	openalex.org	1	Post-Quantum Cryptography Migration for Enterprise Security
SIG-13AAE2CD702E	2026-01-19	MDPI AG	1	Cryptographic Asset Discovery and Inventory for Embedded Systems: A Framework for Post-Quantum ...
SIG-5C7568EBD975	2026-01-01	Elsevier BV	1	The Role of the Second Line of Defense in Post-Quantum Cryptography Migration and Cryptographic...
SIG-D45704585F54	2026-01-01	Elsevier BV	1	Towards a Quantum Risk Index: a risk-based framework for prioritizing Post-Quantum Cryptography...
SIG-CA65D28B8AC6	2025-12-18	openalex.org	1	Harvest-Now, Decrypt-Later: A Temporal Cybersecurity Risk in the Quantum Transition
SIG-80F45F65BE1F	2025-10-13	openalex.org	1	Blockchain Security Risk Assessment in Quantum Era, Migration Strategies, and Proactive Defense
SIG-2153E369E0CC	2025-09-15	Journal of Electrical and...	1	Securing Digital Communications: Quantum and Post-Quantum Cryptography
SIG-43901F3A86B3	2025-04-09	Wiley	1	Post-Quantum Cryptography: Evaluating and Standardizing Quantum-Resistant Algorithms
SIG-2E8BDC1715A1	2026-08-18	Kalkine Media	2	Quantum Secure Encryption Unveils QSim to Streamline Enterprise Post-Quantum Cryptography Migra...
SIG-FB099EF70AC5	2026-07-01	Quantum Computing Report	2	Department of War Unveils Enterprise Post-Quantum Cryptography Strategy to Insulate Defense Net...
SIG-5631159A1ECA	2026-06-23	U.S. Department of War (...)	2	Securing Global Dominance: DoW Unleashes Quantum Defense Strategy to Harden Networks and E - U....
SIG-D352790E1A52	2026-07-21	useekko.app	3	Show HN: Ekko – post-quantum E2EE layer for Instagram, WhatsApp, Telegram Web

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:14+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:17+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:57:26+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.